



BookNest

Library Information Management System

Technical System Manual

For System Administrators, Database Managers & Developers

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Database :MariaDB 10.4.32| Platform: PHP 8.2.12

Executive Summary

BookNest is an enterprise-grade Library Information Management System built on modern, scalable technologies. This comprehensive technical manual provides system administrators, database managers, and software developers with complete documentation of the system architecture, database structure, modules, configuration, and operational procedures.

Key Technical Capabilities:

- Three-tier architecture designed for scalability
- 55 relational database tables with comprehensive data models
- Role-based access control with complete audit logging
- Multi-branch support with centralized database
- Comprehensive reporting and analytics
- Enterprise-grade security and data protection

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1. Technology Stack & System Requirements

1.1 Hardware Requirements

Server Requirements:

- Operating System: Linux (CentOS 7+) or Windows Server 2016+
- Processor: Multi-core processor (minimum 4 cores for production)
- RAM: Minimum 8 GB (16 GB recommended for production)
- Storage: Minimum 100 GB SSD
- Network: Gigabit Ethernet connection

1.2 Software Stack

Component	Version
Web Server	Apache 2.4+ or Nginx 1.19+
PHP	8.2.12 or higher
Database	MariaDB 10.4.32 or higher
Character Set	UTF8MB4

1.3 Browser Support

- Google Chrome 90+
- Mozilla Firefox 88+
- Microsoft Edge 90+
- Safari 14+ (macOS)

2. System Architecture Overview

2.1 Three-Tier Architecture

BookNest implements a three-tier architecture that provides scalability, maintainability, and clear separation of concerns:

Tier	Components
Presentation	HTML/CSS/JavaScript web interface
Business Logic	PHP application logic
Data Layer	MariaDB relational database

2.2 Core Features

- Book inventory management with ISBN tracking
- Circulation and issue/return with automatic penalties
- Journal and periodical management
- Multi-branch support with centralized data
- Comprehensive reporting and analytics
- Role-based access control

3. Installation & Deployment Guide

3.1 Pre-Installation Checklist

- Verify all hardware and software requirements
- Install web server (Apache/Nginx)
- Install PHP 8.2.12 with required extensions
- Install MariaDB 10.4.32
- Configure SSL/TLS certificates
- Plan backup and recovery strategy

3.2 Installation Steps

- Extract BookNest application files to web directory
- Create database with UTF8MB4 character set
- Import database schema

- Configure database connection parameters
- Set file permissions appropriately
- Configure web server virtual host
- Create initial administrator account
- Verify application access

4. Database Architecture & Schema

4.1 Database Overview

Property	Value
Database Name	Booknest (MariaDB 10.4.32)
Total Tables	55
Character Set	UTF8MB4

4.2 Core Table Groups

Master Data Tables:

Table	Purpose
books	Book records
bookinfo	Detailed info
bookcategory	Categories
booksupplier	Suppliers
publisher	Publishers

Circulation Tables:

Table	Purpose
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bookcirculation	Issue/return tracking
librarybook	Physical copies
counter	Desk information
transferstatusbook	Transfers

Journal Tables:

Table	Purpose
journalmaster	Journal records
journalsubscription	Subscriptions
periodicalreceipt	Receipts
journalpublisher	Publishers

User & Access Tables:

Table	Purpose
users	User accounts
student	Student info
staff	Staff info
libroles	Roles
libactivity	Audit logs

Financial Tables:

Table	Purpose
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budgetallocation	Budget details
budgetinfo	Budget info
fundtype	Fund types
penaltypolicy	Penalty config

5. Core System Modules

5.1 Book Management Module

Handles all book inventory management. Key functions: add/update/delete books, categorize with classifications, track editions and suppliers, generate call numbers, manage covers, track copies across branches.

5.2 Circulation Module

Manages borrowing and returns. Key functions: issue books with due dates, process returns, calculate penalties, handle renewals, manage holds, generate receipts and reports.

5.3 Journal & Periodical Module

Manages journal subscriptions. Key functions: create subscriptions, track frequency and receipts, monitor renewals, manage backbound volumes, track distributors.

5.4 Member Management Module

Manages library members. Key functions: register members, assign roles, manage status, track memberships, view borrowing history.

5.5 Administration Module

System configuration. Key functions: configure institution details, set policies, allocate budgets, manage holidays, generate reports.

6. User Management & Security

6.1 User Types & Roles

User Type	Description	Permissions
Admin	System admin	All features

Librarian	Staff	Circulation, cataloging
Staff	Institutional staff	Borrowing
Student	Student members	Borrowing
Visitor	Guest	View-only

6.2 Access Control

- Role-based access control (RBAC)
- Granular permissions per role
- Complete audit logging of changes
- Multiple roles per user support

6.3 Security Best Practices

- Use strong passwords (minimum 12 characters)
- Enable SSL/TLS for all connections
- 30-minute inactivity timeout
- 90-day password change enforcement
- Regular access rights review
- Monitor activity logs

7. Library Operations & Workflows

7.1 Book Circulation Workflow

- Member presents book and ID
- Librarian searches for book
- System verifies member eligibility
- System checks borrowing limit
- System checks for pending penalties
- System calculates due date

- Issue recorded and receipt generated

7.2 Penalty Calculation

Penalties calculated automatically: Penalty Amount equals (ReturnDate minus DueDate) multiplied by Daily Penalty Rate. Rates vary by member type, with configurable maximum caps.

7.3 Stock Management

- Physical inventory verification
- Database reconciliation
- Inter-branch transfer tracking
- Quarterly inventory reports
- Discrepancy identification

8. Backup & Recovery Procedures

8.1 Backup Strategy

- Daily full database backups
- Hourly incremental backups during business hours
- Weekly off-site replication
- Monthly archive backups for compliance

8.2 Recovery Procedures

- Identify backup needed
- Stop application
- Create fresh or empty database
- Restore from backup
- Verify data integrity
- Restart and test

9. Performance Optimization

9.1 Database Optimization

- Create indexes on frequently queried columns

- Regular database maintenance
- Archive old circulation records
- Optimize query patterns

9.2 Application Optimization

- Enable PHP opcode caching
- Implement query result caching
- Use persistent connections
- Compress assets
- Enable GZIP compression

10. Troubleshooting Guide

10.1 Common Issues

Book Cannot Be Issued:

- Check member status (must be Active)
- Verify borrowing limit not exceeded
- Check for pending penalties
- Ensure book exists in system

Incorrect Penalty Calculations:

- Verify policy configuration
- Check date accuracy
- Confirm member type

Database Connection Errors:

- Verify MariaDB is running
- Check configuration parameters
- Confirm user permissions
- Verify network connectivity

11. Complete Data Dictionary

11.1 Books Table Schema

Column Name	Data Type	Description
BookID	BIGINT	Primary key for book
BookTitle	VARCHAR(255)	Book title
BookAuthor	VARCHAR(255)	Author name
BookStatus	VARCHAR(20)	Active/Inactive status
ISBN	VARCHAR(20)	ISBN number

11.2 Bookcirculation Table Schema

Column Name	Data Type	Description
BCID	BIGINT	Circulation record ID
UserID	BIGINT	Borrower ID
IssueDate	DATE	Issue date
DueDate	DATE	Due date
ReturnDate	DATE	Return date

11.3 Users Table Schema

Column Name	Data Type	Description
UserID	BIGINT	User ID
UserName	VARCHAR(50)	Login name
UserStatus	VARCHAR(20)	Status

DateOfEnroll	DATE	Enrollment date
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12. Development & Customization

12.1 Directory Structure

/booknest/ contains: config/ (database settings), classes/ (PHP classes), modules/ (feature modules), reports/ (reporting), api/ (REST endpoints), templates/ (HTML), assets/ (CSS/JS/images).

12.2 Adding Custom Fields

- Identify target table
- Create database migration
- Update PHP classes
- Modify form templates
- Add validation
- Update reports
- Test thoroughly

12.3 Database Migration Best Practices

- Always backup before migrations
- Test on development first
- Schedule during low traffic
- Document all changes
- Maintain rollback scripts
- Verify data integrity

13. Frequently Asked Questions

Q: How do I add a new book to the system?

A: Access Book Management module, click Add New Book, fill in title, author, ISBN, select category and publisher, then submit.

Q: Can I modify due dates for issued books?

A: Yes, librarians can extend due dates through circulation module. This updates DueDate and recalculates penalties.

Q: How are penalties calculated?

A: Automatically when books are returned late: (ReturnDate minus DueDate) multiplied by daily penalty rate.

Q: Can I track multiple book copies?

A: Yes, each copy is a separate record in librarybook table, allowing tracking across branches.

Q: What backup frequency is recommended?

A: Daily full backups, hourly incremental during business hours, weekly off-site replication.

Q: How are user roles managed?

A: Roles are in libroles table. Administrators assign through Admin Panel. All changes are logged.

14. Glossary of Technical Terms

API - Application Programming Interface for external system integration

BCID - Book Circulation ID, unique for each circulation record

CRUD - Create, Read, Update, Delete database operations

ISBN - International Standard Book Number

MariaDB - Open-source relational database management system

RBAC - Role-Based Access Control for permission management

UTF8MB4 - Unicode character encoding supporting all languages

End of Technical Manual

For additional support, contact your system administrator or refer to online documentation.

[BookNest Library Management System v1.0](#)